BT401 – Mathematics-III

Unit – I (Numerical Methods – I)

- **Main**
  - Algebraic Equations
  - Finite Differences
  - Interpolation
- **Detail**
  - Polynomial & transcendental equations
  - Bisection – Newton Raphson – Regula Falsi
  - Finite difference operators – relations
  - Newton forward – backward interpolation
  - Newton divided difference
  - Lagrange interpolation

Unit – II (Numerical Methods – II)

- **Main**
  - Numerical Differentiation
  - Numerical Integration
  - Linear Equations
- **Detail**
  - Numerical differentiation
  - Trapezoidal rule
  - Simpson's 1/3 – 3/8 rules
  - Gauss elimination – Gauss Jordan – Crout
  - Jacobi – Gauss Seidel – Relaxation

Unit – III (Numerical Methods – III)

- **Main**
  - Ordinary Differential Equations
  - Predictor Corrector Methods
  - Partial Differential Equations

- **Detail**
    - Taylor – Euler – Modified Euler
    - Runge Kutta fourth order
    - Milne's – Adam's methods
    - Laplace equation
    - Poisson equation
    - Heat equation (Bender–Schmidt – Crank–Nicholson)
    - Wave equation

Unit – IV (Transform Calculus)

- **Main**
    - Laplace Transform
    - Inverse Laplace
    - Fourier Transform
- **Detail**
    - Properties of Laplace transform
    - Laplace of periodic functions
    - Inverse Laplace methods
    - Convolution theorem
    - Evaluation of integrals
    - Solving ODEs
    - Fourier transform

Unit – V (Probability & Statistics)

- **Main**
    - Probability Functions
    - Discrete Distributions
    - Continuous Distributions
- **Detail**
    - Probability mass function
    - Probability density function
    - Binomial distribution
    - Poisson distribution

- Normal distribution
- Exponential distribution

CS402 – Analysis & Design of Algorithms

Unit – I

- **Main**
  - Algorithm Basics
  - Asymptotic Notations
  - Divide & Conquer
- **Detail**
  - Designing – analyzing algorithms
  - Heap – heap sort
  - Binary search
  - Merge sort
  - Quick sort
  - Strassen's matrix multiplication

Unit – II

- **Main**
  - Greedy Method
  - Optimization Problems
  - Shortest Path
- **Detail**
  - Optimal merge pattern
  - Huffman coding
  - Minimum spanning tree
  - Knapsack problem
  - Job sequencing
  - Single source shortest path

Unit – III

- **Main**

    - Dynamic Programming
    - Graph Algorithms
    - Reliability Design
- **Detail**
    - 0/1 knapsack
    - Multistage graph
    - Reliability design
    - Floyd-Warshall algorithm

Unit – IV

- **Main**
    - Backtracking
    - Branch & Bound
    - Parallel Algorithms
- **Detail**
    - 8-Queen problem
    - Hamiltonian cycle
    - Graph coloring
    - Traveling salesman problem
    - Lower bound theory

Unit – V

- **Main**
    - Tree Structures
    - Graph Traversals
    - NP Completeness
- **Detail**
    - Binary search trees
    - Height balanced trees
    - 2-3 trees – B-trees
    - Inorder – Preorder – Postorder
    - DFS – BFS
    - NP-complete problems

CS403 – Software Engineering

Unit – I

- **Main**
  - Software Product
  - Software Process
  - Process Models
- **Detail**
  - Product characteristics
  - Linear sequential model
  - Prototyping – RAD
  - Incremental – Spiral – RUP
  - Agile processes
  - CMM – Process metrics

Unit – II

- **Main**
  - Requirements
  - Requirement Analysis
  - Specification
- **Detail**
  - Functional – Non-functional requirements
  - Requirement elicitation
  - Use case modeling
  - SRS
  - Validation – Traceability

Unit – III

- **Main**
  - Software Design
  - UML
  - Architecture

- **Detail**
    - Design principles
    - Software modeling
    - Architectural styles
    - UI design
    - Component based design
    - Design metrics

Unit – IV

- **Main**
    - Software Testing
    - Testing Techniques
    - OO Analysis
- **Detail**
    - Static – Dynamic analysis
    - Black box – White box testing
    - Unit – Integration – System testing
    - Test cases – Test plan
    - Testing tools

Unit – V

- **Main**
    - Software Maintenance
    - Configuration Management
    - Project Management
- **Detail**
    - Types of maintenance
    - SCM – Version control
    - Re-engineering – Reverse engineering
    - Project planning – Scheduling
    - Risk management
    - SQA

CS404 – Computer Organization & Architecture

Unit – I

- **Main**
  - Computer Structure
  - CPU Organization
  - Control Unit
- **Detail**
  - Registers – ALU – Bus
  - Instruction formats
  - Addressing modes
  - Hardwired – Microprogrammed control

Unit – II

- **Main**
  - Computer Arithmetic
  - I/O Organization
  - Data Transfer
- **Detail**
  - Signed arithmetic
  - Booth algorithm
  - DMA
  - Serial – Parallel transfer

Unit – III

- **Main**
  - Memory Organization
  - Cache Memory
  - Virtual Memory
- **Detail**

    - RAM – ROM – Secondary memory
    - Cache mapping
    - Replacement algorithms
    - Memory management hardware

Unit – IV

- **Main**
    - Multiprocessors
    - Pipelining
    - Vector Processing
- **Detail**
    - Inter-processor communication
    - Synchronization
    - RISC – CISC
    - Multicore processors

Unit – V

- **Main**
    - Advanced Architectures
    - Parallelism
    - Modern Processors
- **Detail**
    - Array processing
    - Vector processing
    - Intel – AMD multicore

CS405 – Operating Systems

Unit – I

- **Main**
  - OS Introduction
  - OS Services
  - System Calls
- **Detail**
  - OS evolution – types
  - Utility programs
  - System call mechanism

Unit – II

- **Main**
  - File Systems
  - Disk Management
  - Directory Structure
- **Detail**
  - File concepts
  - Disk allocation methods
  - Disk scheduling algorithms

Unit – III

- **Main**
  - CPU Scheduling
  - Memory Management
  - Virtual Memory
- **Detail**
  - Scheduling algorithms
  - Paging – Segmentation
  - Demand paging

Unit – IV

- **Main**
  - I/O Management
  - Concurrency
  - Deadlocks
- **Detail**
  - Interrupt driven I/O
  - Semaphores
  - Deadlock prevention – recovery

Unit – V

- **Main**
  - Distributed OS
  - Network OS
  - Case Studies
- **Detail**
  - Unix / Linux
  - Windows
  - Contemporary OS

CS406 (A) – Programming Practices using Java

Unit – I

- **Main**
  - Java Basics
  - Data Types
  - Operators
- **Detail**
  - Java features – JVM – JRE – JDK
  - Variables – literals – keywords
  - Primitive – non-primitive data types
  - Arithmetic – relational – logical operators

Unit – II

- **Main**
  - Control Statements
  - Arrays
  - Strings
- **Detail**
  - If – switch – loops
  - One dimensional – multidimensional arrays
  - String – StringBuffer – StringBuilder

Unit – III

- **Main**
  - OOP Concepts
  - Classes
  - Objects
- **Detail**
  - Encapsulation – inheritance – polymorphism – abstraction
  - Class definition – constructors
  - Method overloading – overriding

Unit – IV

- **Main**
  - Exception Handling
  - Multithreading
  - Packages
- **Detail**

    - Try – catch – finally
    - User defined exceptions
    - Thread lifecycle – synchronization
    - Packages – interfaces

Unit – V

- **Main**
    - File Handling
    - Applets
    - GUI
- **Detail**
    - File input – output streams
    - Applet lifecycle
    - AWT – Swing – Event handling

CS406 (B) – Programming Practices using .NET

Unit – I

- **Main**
  - .NET Framework
  - CLR
  - C# Basics
- **Detail**
  - .NET architecture
  - Common Language Runtime
  - Variables – data types – operators

Unit – II

- **Main**
  - Control Structures
  - Arrays
  - Strings
- **Detail**
  - Selection – iteration statements
  - Arrays – collections
  - String handling methods

Unit – III

- **Main**
  - OOP in C#
  - Classes
  - Interfaces
- **Detail**
  - Encapsulation – inheritance – polymorphism
  - Abstract classes
  - Interfaces – namespaces

Unit – IV

- **Main**
  - Exception Handling
  - Delegates
  - Events
- **Detail**
  - Try – catch – finally

    - Multicast delegates
    - Event driven programming

Unit – V

- **Main**
    - File Handling
    - ASP.NET
    - Database Connectivity
- **Detail**
    - File streams
    - ASP.NET basics
    - ADO.NET – database access

CS406 (C) – Programming Practices using Python

Unit – I

- **Main**
  - Python Basics
  - Data Types
  - Operators
- **Detail**
  - Python features
  - Variables – keywords
  - Numeric – string – list – tuple – dictionary
  - Arithmetic – logical – membership operators

Unit – II

- **Main**
  - Control Statements
  - Functions
  - Modules
- **Detail**
  - If – loops
  - User defined functions
  - Lambda functions
  - Importing modules

Unit – III

- **Main**
  - Strings
  - Lists
  - Tuples
- **Detail**
  - String operations
  - List methods
  - Tuple operations

Unit – IV

- **Main**
  - Dictionaries
  - File Handling
  - Exception Handling

- **Detail**
    - Dictionary methods
    - Read – write files
    - Try – except – finally

Unit – V

- **Main**
    - OOP in Python
    - GUI
    - Database Connectivity
- **Detail**
    - Classes – objects – inheritance
    - Tkinter
    - Python with database

CS406 (D) – Programming Practices using MATLAB

Unit – I

- **Main**
  - MATLAB Environment
  - Variables
  - Operators
- **Detail**
  - MATLAB features
  - Command window – editor
  - Arithmetic – relational operators

Unit – II

- **Main**
  - Arrays
  - Matrices
  - Control Statements
- **Detail**
  - Vector operations
  - Matrix manipulation
  - If – for – while

Unit – III

- **Main**
  - Functions
  - Scripts
  - File Handling
- **Detail**
  - User defined functions
  - Script files
  - Data files

Unit – IV

- **Main**
  - Plotting
  - Graphics
  - Visualization
- **Detail**
  - 2D – 3D plots

    - Graph customization
    - Subplots

Unit – V

- **Main**
    - Toolboxes
    - Simulink
    - Applications
- **Detail**
    - Signal processing toolbox
    - Simulink models
    - Engineering applications